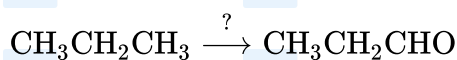


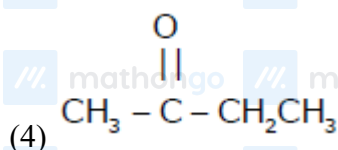
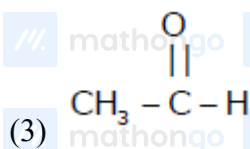
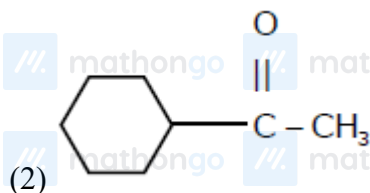
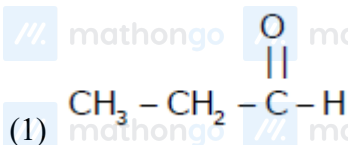
Q1: 24 Feb (Shift 1) - Single Correct

Which of the following reagent is used for the following reaction?

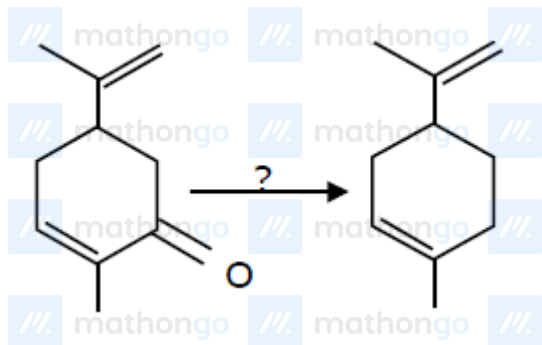


- (1) Potassium permanganate
- (2) Molybdenum oxide
- (3) Copper at high temperature and pressure
- (4) Manganese acetate

Q2: 24 Feb (Shift 2) - Single Correct

Which one of the following carbonyl compounds cannot be prepared by addition of water on an alkyne in the presence of HgSO_4 and H_2SO_4 ?

Q3: 24 Feb (Shift 2) - Single Correct



Which of the following reagent is suitable for the preparation of the product in the above reaction.

(1) Red P + Cl₂

(2) $\text{NH}_2\text{-NH}_2 / \text{C}_2\text{H}_5\text{ONa}^{\ominus\oplus}$

(3) Ni/H₂

(4) NaBH₄

Q4: 25 Feb (Shift 1) - Single Correct

Given below are two statements:

Statement-I : CeO₂ can be used for oxidation of aldehydes and ketones.

Statement-II : Aqueous solution of EuSO₄ is a strong reducing agent.

(1) Statement I is true, statement II is false

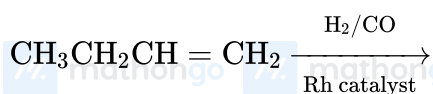
(2) Statement I is false, statement II is true

(3) Both Statement I and Statement II are false

(4) Both Statement I and Statement II are true

Q5: 25 Feb (Shift 2) - Single Correct

The major product of the following reaction is :

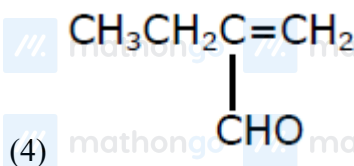


(1) CH₃CH₂CH₂CHO

(2) CH₃CH₂CH=CH-CHO

Questions with Answer Keys

MathonGo



Q6: 26 Feb (Shift 2) - Single Correct

2,4 -DNP test can be used to identify:

(1) aldehyde

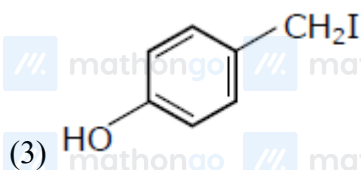
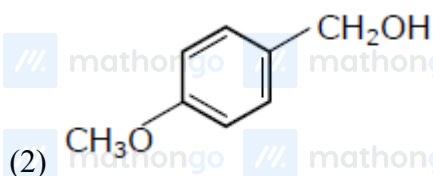
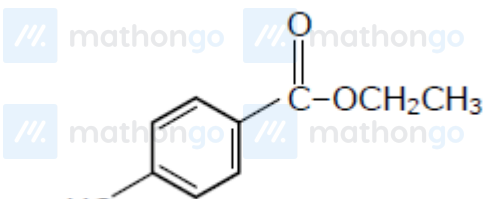
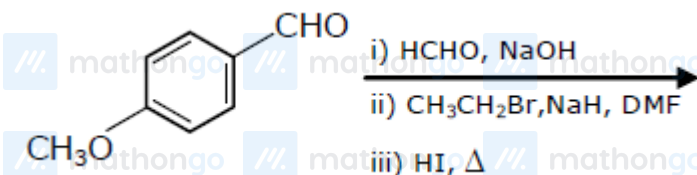
(2) halogens

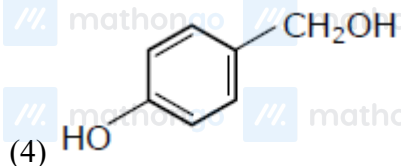
(3) ether

(4) amine

Q7: 26 Feb (Shift 2) - Single Correct

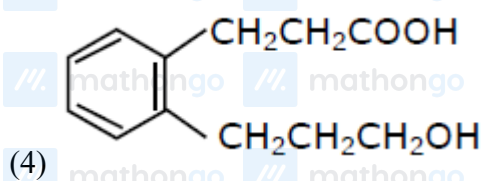
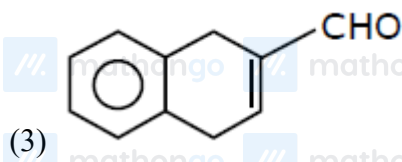
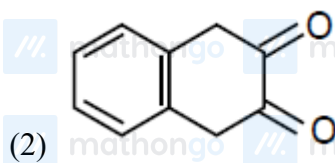
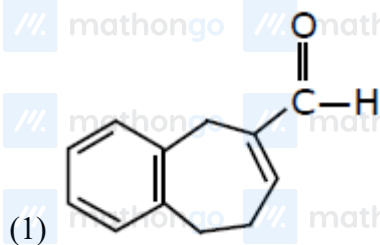
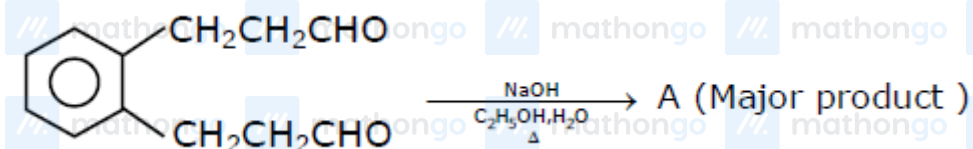
Identify A in the following chemical reaction.



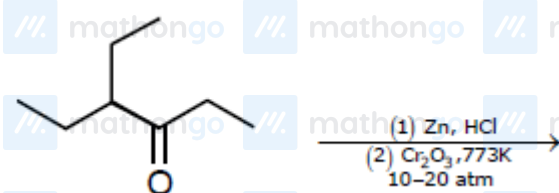


Q8: 26 Feb (Shift 2) - Single Correct

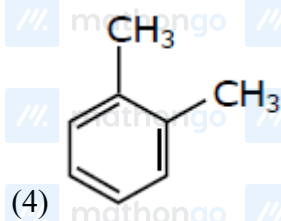
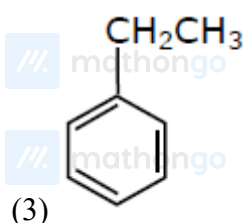
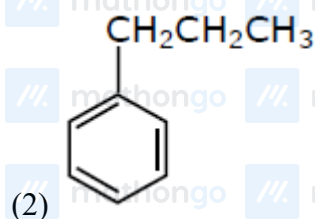
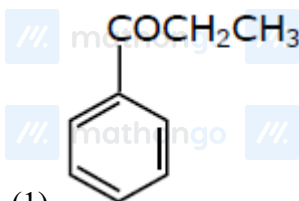
Identify A in the given chemical reaction.



Q9: 26 Feb (Shift 2) - Single Correct



Considering the above reaction, the major product among the following is:

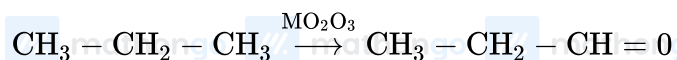


Answer Key**Q1 (2)****Q2 (1)****Q3 (2)****Q4 (4)****Q5 (3)****Q6 (1)****Q7 (3)****Q8 (1)****Q9 (3)**

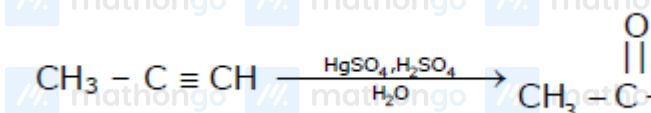
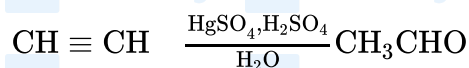
Hints and Solutions

MathonGo

Q1 (2)

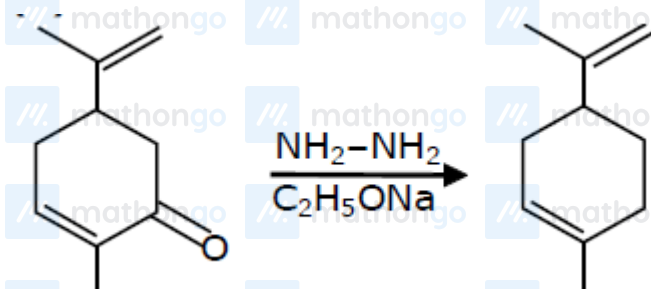


Q2 (1)

Reaction of Alkyne with HgSO_4 & H_2SO_4 follow asHence, by this process preparation of $\text{CH}_3\text{CH}_2\text{CHO}$

Can't possible.

Q3 (2)

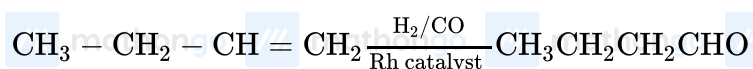


It is wolf-kishner reduction of carbonyl compounds.

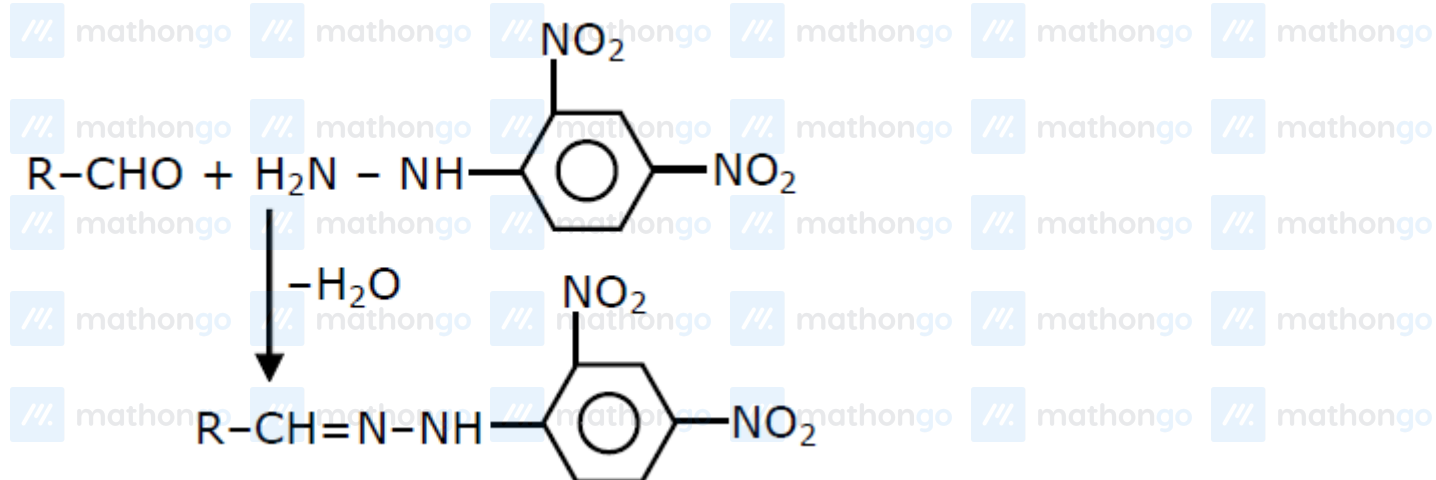
Q4 (4)

 CeO_2 can be used as oxidising agent like seO_2 . Similarly EuSO_4 used as a reducing agent.

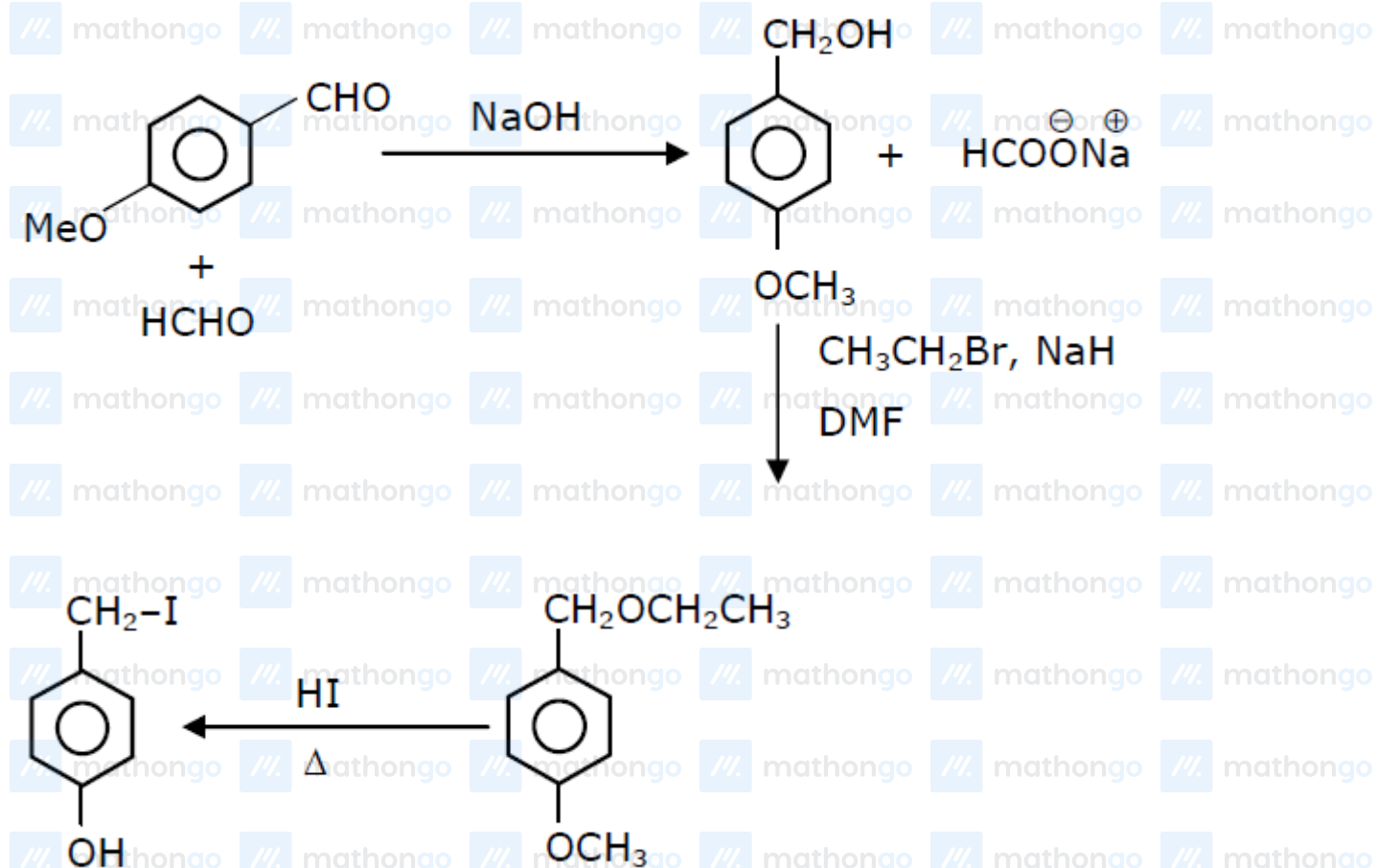
Q5 (3)



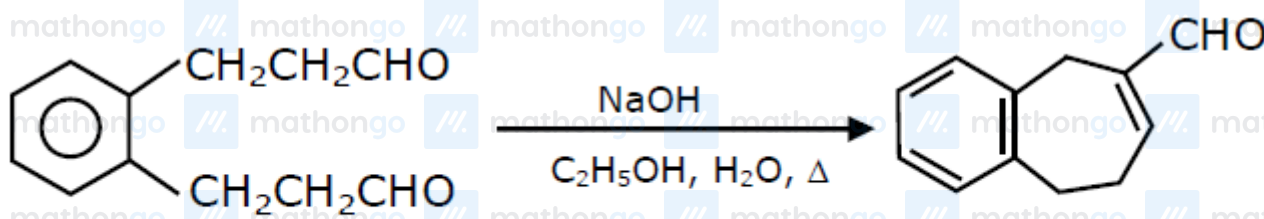
Q6 (1)



Q7 (3)

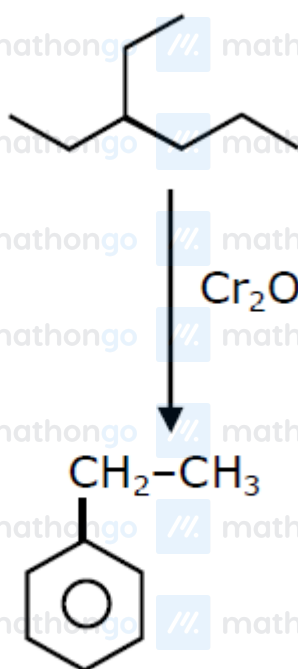
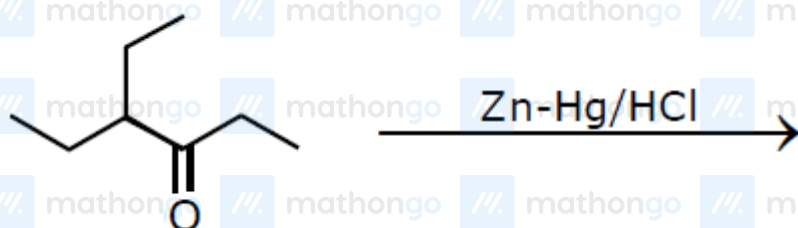


Q8 (1)



(Internal aldol condensation)

Q9 (3)



(Aromatisation)